Heat Kernels and Index Theory

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Contents

List of Figures

Chapter 1

Geometric preliminaries

Throughout, M is a smooth compact manifold without boundary unless stated otherwise. All vector bundles are finite-dimensional and smooth.

1.1 Connections and curvature

Definition 1.1. A connection on a real or complex vector bundle $E \to M$ is an $\mathbb{R}$-linear map

$$\nabla : \Gamma(E) \longrightarrow \Omega^1(M; E)$$

satisfying

$$\nabla(fs) = df \otimes s + f\nabla s.$$

For a vector field X, write $\nabla_X s = (\nabla s)(X)$.

In a local frame $e = (e_1, \dots, e_r)$, a section is a column vector and

$$\nabla = d + \omega, \qquad \omega \in \Omega^1(U; \operatorname{End} E).$$

If the frame changes by $e' = eg$, then

$$\omega' = g^{-1}\omega g + g^{-1}dg.$$

This transformation law is exactly what makes the definition independent of a frame.

The connection extends to E-valued forms by

$$d_\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^{\deg \alpha}\alpha \wedge \nabla s.$$

Definition 1.2. The curvature is

$$F^\nabla = d_\nabla^2 \in \Omega^2(M; \operatorname{End} E).$$

In a local frame,

$$F^\nabla = d\omega + \omega \wedge \omega.$$

Because d_∇^2 is $C^\infty(M)$-linear, it is an endomorphism-valued two-form. The graded Jacobi identity gives the Bianchi identity

$$d_\nabla F^\nabla = 0.$$

Question 1.3. *Why is a connection not tensorial while its curvature is?*

Answer. The Leibniz rule contains the derivative df, so $\nabla(fs) \neq f\nabla s$. Applying the connection twice makes the first-derivative terms cancel:

$$d_\nabla^2(fs) = d_\nabla(df \otimes s + f\nabla s) = -df \wedge \nabla s + df \wedge \nabla s + f\, d_\nabla^2 s.$$

Thus $d_\nabla^2(fs) = f\, d_\nabla^2 s$. This calculation is the intrinsic reason the inhomogeneous term $g^{-1}dg$ in the transformation law for ω disappears from the transformation law $F' = g^{-1}Fg$. □

If E is Hermitian, a connection is unitary when

$$X\langle s_1, s_2\rangle = \langle \nabla_X s_1, s_2\rangle + \langle s_1, \nabla_X s_2\rangle.$$

In a unitary frame, its connection form is skew-Hermitian and its curvature is skew-Hermitian.

1.2 The Levi–Civita connection

Let g be a Riemannian metric. A connection on TM is metric if $\nabla g = 0$, and its torsion is

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y].$$

Theorem 1.4 (Levi–Civita)**.** *There is a unique metric connection on TM with zero torsion. It is determined by the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y,Z) + Yg(Z,X) - Zg(X,Y) \\ & - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Proof. Metric compatibility expresses $Xg(Y,Z)$, $Yg(Z,X)$, and $Zg(X,Y)$ through covariant derivatives. Add the first two expressions, subtract the third, and use zero torsion to obtain the displayed formula. It proves uniqueness.

For existence, denote the right-hand side of the Koszul formula by $K(X,Y,Z)$. The derivatives of a function multiplying Z cancel, so $K(X,Y,fZ) = fK(X,Y,Z)$. Nondegeneracy of g therefore defines a unique vector field $\nabla_X Y$ by $2g(\nabla_X Y, Z) = K(X,Y,Z)$. Expanding the same formula gives

$$K(fX,Y,Z) = fK(X,Y,Z), \qquad K(X,fY,Z) = fK(X,Y,Z) + 2X(f)g(Y,Z).$$

Hence $\nabla_{fX} Y = f\nabla_X Y$ and $\nabla_X(fY) = X(f)Y + f\nabla_X Y$, so this is a connection. Finally,

$$K(X,Y,Z) - K(Y,X,Z) = 2g([X,Y],Z)$$

proves that its torsion vanishes, while

$$K(X,Y,Z) + K(X,Z,Y) = 2Xg(Y,Z)$$

proves metric compatibility. Thus the connection defined by the formula has both required properties. □

In local coordinates $x^1, \dots, x^n$, write $\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k$. Substitution in the Koszul formula gives

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell}\big(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\big).$$

This both recovers the coordinate calculation in the original lectures and shows why geodesic normal coordinates satisfy $g_{ij}(p) = \delta_{ij}$ and $\Gamma^k_{ij}(p) = 0$.

The Riemann curvature tensor is

$$R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

In the same convention,

$$R(\partial_i,\partial_j)\partial_k = R^\ell{}_{kij}\partial_\ell, \qquad R^\ell{}_{kij} = \partial_i\Gamma^\ell_{jk} - \partial_j\Gamma^\ell_{ik} + \Gamma^m_{jk}\Gamma^\ell_{im} - \Gamma^m_{ik}\Gamma^\ell_{jm}.$$

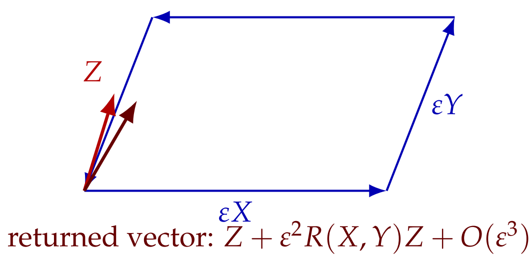

Figure 1.1: Curvature is the second-order failure of parallel transport around an infinitesimal parallelogram to return a vector unchanged.

With $R(X,Y,Z,W) = g(R(X,Y)Z,W)$, it satisfies

$$\begin{aligned} R(X,Y,Z,W) &= -R(Y,X,Z,W), \\ R(X,Y,Z,W) &= -R(X,Y,W,Z), \\ R(X,Y,Z,W) &= R(Z,W,X,Y), \end{aligned}$$

together with the first Bianchi identity

$$R(X,Y)Z + R(Y,Z)X + R(Z,X)Y = 0.$$

The Ricci tensor and scalar curvature are the contractions

$$\operatorname{Ric}(X,Y) = \sum_j g(R(e_j,X)Y,e_j), \qquad \operatorname{scal} = \sum_j \operatorname{Ric}(e_j,e_j).$$

1.3 Chern–Weil forms

Let f be an invariant polynomial on a matrix Lie algebra. The form $f(F^\nabla)$ is closed, and its de Rham class does not depend on the connection.

Proposition 1.5 (Transgression). *For a smooth family ∇^t with curvature F_t,*

$$\frac{d}{dt} f(F_t) = d\big(k\, f(\dot\nabla^t, F_t, \dots, F_t)\big)$$

when f is homogeneous of degree k.

Proof. The difference of two connections is tensorial; consequently $\alpha_t := \dot{\nabla}^t$ is a globally defined element of $\Omega^1(M; \operatorname{End} E)$. In a local frame $\nabla^t = d + \omega_t$, it is represented by $\alpha_t = \dot{\omega}_t$. Since

$$F_t = d\omega_t + \omega_t \wedge \omega_t,$$

differentiation gives

$$\dot{F}_t = d\alpha_t + \omega_t \wedge \alpha_t + \alpha_t \wedge \omega_t = d_{\nabla^t}\alpha_t.$$

Polarize f to a symmetric k-linear invariant form. Then

$$\frac{d}{dt} f(F_t, \dots, F_t) = k\, f(d_{\nabla^t}\alpha_t, F_t, \dots, F_t).$$

For endomorphism-valued forms, invariance of f implies the covariant Leibniz identity

$$d\, f(B_1, \dots, B_k) = \sum_j (-1)^{\deg B_1 + \cdots + \deg B_{j-1}} f(B_1, \dots, d_{\nabla^t} B_j, \dots, B_k);$$

the commutator terms cancel inside f. Apply this with $B_1 = \alpha_t$ and $B_2 = \cdots = B_k = F_t$. The Bianchi identity $d_{\nabla^t} F_t = 0$ kills every term except the first, proving

$$\frac{d}{dt} f(F_t) = d\big(k f(\alpha_t, F_t, \dots, F_t)\big).$$

Integrating from $t = 0$ to 1 shows explicitly that two connections produce cohomologous characteristic forms. $\square$

Discussion. Chern–Weil theory is the bridge between analysis and topology used later. The heat kernel produces a particular curvature polynomial. Transgression shows that its integral on a closed manifold is independent of the connection used to compute it.

For a complex bundle E, the Chern character form is

$$\operatorname{ch}(E, \nabla) = \operatorname{Tr} \exp\left(\frac{iF^{\nabla}}{2\pi}\right).$$

For a real oriented bundle with curvature matrix R, define

$$\widehat{A}(E, \nabla) = \det{}^{1/2}\left(\frac{R/(4\pi i)}{\sinh(R/(4\pi i))}\right),$$

and

$$L(E, \nabla) = \det{}^{1/2}\left(\frac{R/(2\pi i)}{\tanh(R/(2\pi i))}\right).$$

These power-series expressions are interpreted through invariant polynomials. Their cohomology classes are denoted $\operatorname{ch}(E)$, $\widehat{A}(E)$, and $L(E)$.

The determinant notation is shorthand for traces of power series. If $f(z) = 1 + O(z)$, then

$$\det f(F) = \exp\big(\operatorname{Tr} \log f(F)\big), \qquad \det{}^{1/2} f(F) = \exp\left(\frac{1}{2} \operatorname{Tr} \log f(F)\right).$$

Although these are written as infinite series, only finitely many terms can contribute in any fixed differential-form degree. Conjugating the curvature does not change its traces, so every coefficient is an invariant polynomial and Chern–Weil theory applies.

Question 1.6. *Why do traces and determinants appear in the index density?*

Answer. Curvature is an endomorphism-valued two-form, whereas the index density must be an ordinary scalar form. The trace turns powers of curvature into scalar forms and is additive under direct sum. Exponentiating traces packages the multiplicative characteristic sequences, such as $\widehat{A}$ and L, that arise from independent curvature two-plane blocks in the heat model. □

For a complex bundle with Chern roots x_j, the Todd form is

$$\mathrm{Td}(E) = \prod_j \frac{x_j}{1 - e^{-x_j}}.$$

The splitting principle makes this a well-defined polynomial in the Chern classes.

1.4 Differential complexes

A differential complex is a sequence

$$0 \longrightarrow \Gamma(E^0) \xrightarrow{D_0} \Gamma(E^1) \xrightarrow{D_1} \cdots \xrightarrow{D_{r-1}} \Gamma(E^r) \longrightarrow 0, \qquad D_{j+1}D_j = 0.$$

Its cohomology is

$$H^j(D) = \ker D_j / \operatorname{im} D_{j-1}.$$

When these spaces are finite-dimensional, its Euler characteristic is

$$\chi(D) = \sum_j (-1)^j \dim H^j(D).$$

The basic example is the de Rham complex

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \longrightarrow 0.$$

Its cohomology is the de Rham cohomology of M, and its Euler characteristic is the topological Euler characteristic $\chi(M)$.

Chapter 2

Elliptic operators and index

2.1 Differential operators and principal symbols

A differential operator $P : \Gamma(E) \to \Gamma(F)$ of order at most m has the local form

$$P = \sum_{|\alpha| \le m} a_\alpha(x)\partial_x^\alpha.$$

Its principal symbol is

$$\sigma_m(P)(x,\xi) = \sum_{|\alpha|=m} a_\alpha(x)(i\xi)^\alpha : E_x \longrightarrow F_x.$$

Equivalently,

$$\sigma_m(P)(x,df)s = \lim_{\tau\to\infty} \tau^{-m} e^{-i\tau f} P(e^{i\tau f}s)(x).$$

The latter formula proves coordinate invariance.

Discussion. The oscillatory test $e^{i\tau f}s$ isolates the part of P in which all m derivatives fall on the phase. Derivatives of the section, the coefficients, or lower-order terms carry fewer powers of τ and disappear after multiplication by τ^{-m}. This is why the principal symbol is intrinsic although the coefficients a_α are not.

The symbol is multiplicative:

$$\sigma_{m+\ell}(PQ) = \sigma_m(P)\sigma_\ell(Q).$$

If metrics and a density are fixed, the formal adjoint satisfies

$$\sigma_m(P^*)(x,\xi) = \sigma_m(P)(x,\xi)^*.$$

Formal adjoints and the Hodge star

The formal-adjoint discussion in the original lectures is used repeatedly later, so we record it rather than hiding it in the symbol formula. Fix Hermitian metrics on E, F and a positive smooth density on M. There is a unique differential operator

$$P^* : C^\infty(M;F) \longrightarrow C^\infty(M;E)$$

such that

$$\langle Pu, v\rangle_{L^2} = \langle u, P^*v\rangle_{L^2}$$

for all smooth sections. In a coordinate chart, repeated integration by parts moves each ∂^α from u to the product of the coefficient, density, and v. A partition of unity patches the result; uniqueness follows from nondegeneracy of the L^2-pairing. Moving all m derivatives produces complex conjugation and the factor $(-1)^m$; with the convention $\sigma_m(P)(x,\xi) = \sum a_\alpha (i\xi)^\alpha$, these combine exactly to give the displayed adjoint-symbol identity.

If M is oriented, the Hodge star is the complex-linear map

$$* : \Lambda^k T^* M_{\mathbb{C}} \longrightarrow \Lambda^{n-k} T^* M_{\mathbb{C}}$$

characterized by

$$\alpha \wedge *\overline{\beta} = \langle \alpha, \beta \rangle \, d\,\mathrm{vol}\,.$$

It satisfies $*^2 = (-1)^{k(n-k)}$ on k-forms and

$$d^* = (-1)^{n(k+1)+1} * d * \quad \text{on } \Omega^k(M).$$

Consequently

$$\sigma_1(d)(x,\xi) = i\,\varepsilon(\xi), \qquad \sigma_1(d^*)(x,\xi) = -i\,\iota_{\xi^\sharp},$$

and the de Rham operator has symbol

$$\sigma_1(d+d^*)(x,\xi) = i\big(\varepsilon(\xi) - \iota_{\xi^\sharp}\big), \qquad \sigma_1(d+d^*)^2 = |\xi|^2 I.$$

Question 2.1. *Which parts of this construction require the metric and orientation?*

Answer. The exterior derivative needs neither. The metric and density define the formal adjoint; orientation identifies the density with a volume form and thereby defines the Hodge star. The Laplacian and harmonic representatives therefore depend on the chosen metric, whereas their cohomology classes and the resulting Euler characteristic do not. □

Definition 2.2. The operator P is elliptic if $\sigma_m(P)(x,\xi)$ is invertible for every $\xi \neq 0$.

For a first-order complex $(E^\bullet, D_\bullet)$, the symbol sequence at $(x,\xi) \in T^*M \setminus 0$ is

$$0 \longrightarrow E^0_x \xrightarrow{\sigma(D_0)(x,\xi)} E^1_x \longrightarrow \cdots \longrightarrow E^r_x \longrightarrow 0.$$

The complex is elliptic when this sequence is exact for every nonzero ξ.

Choose bundle metrics. The associated Laplacians are

$$\Delta_j = D_j^* D_j + D_{j-1} D_{j-1}^*.$$

Proposition 2.3. *A complex is elliptic if and only if every Δ_j is elliptic.*

Proof. At a nonzero covector, put $a_j = \sigma(D_j)$. Since $a_j a_{j-1} = 0$,

$$\sigma(\Delta_j) = a_j^* a_j + a_{j-1} a_{j-1}^*.$$

Its quadratic form is

$$\langle \sigma(\Delta_j) v, v \rangle = \|a_j v\|^2 + \|a_{j-1}^* v\|^2.$$

The kernel is therefore $\ker a_j \cap (\operatorname{im} a_{j-1})^\perp$, which vanishes exactly when $\ker a_j = \operatorname{im} a_{j-1}$. □

2.2 Pseudodifferential parametrices

A classical pseudodifferential operator of order m has, in local coordinates, an oscillatory representation

$$Pu(x) = (2\pi)^{-n} \int e^{i(x-y)\cdot\xi} p(x,\xi) u(y)\, dy\; d\xi,$$

modulo a smooth kernel, where

$$p(x,\xi) \sim p_m(x,\xi) + p_{m-1}(x,\xi) + \cdots$$

and p_{m-j} is homogeneous of degree $m-j$ for large ξ. The composition formula is

$$p\#q \sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha p\, D_x^\alpha q, \qquad D_x = \frac{1}{i}\partial_x.$$

Theorem 2.4 (Elliptic parametrix). *If $P \in \Psi^m(M;E,F)$ is elliptic, there is $Q \in \Psi^{-m}(M;F,E)$ such that*

$$QP = I - R_L, \qquad PQ = I - R_R, \qquad R_L, R_R \in \Psi^{-\infty}.$$

Proof. Set $q_{-m} = p_m^{-1}$ outside a neighborhood of the zero section. Suppose $q_{-m}, \ldots, q_{-m-j+1}$ have been chosen. The component of degree $-j$ in $q\#p - I$ is

$$q_{-m-j} p_m + \sum_{\substack{k+\ell+|\alpha|=j \\ k<j}} \frac{1}{\alpha!} \partial_\xi^\alpha q_{-m-k}\, D_x^\alpha p_{m-\ell}.$$

Therefore set

$$q_{-m-j} = -\left(\sum_{\substack{k+\ell+|\alpha|=j \\ k<j}} \frac{1}{\alpha!} \partial_\xi^\alpha q_{-m-k}\, D_x^\alpha p_{m-\ell} \right) p_m^{-1}.$$

This removes the next homogeneous error. To globalize, choose coordinate charts and bundle trivializations, quantize the local asymptotic inverses, and insert a partition of unity on both sides of each local operator. The principal parts agree on overlaps because they are determined intrinsically by p_m^{-1}; hence the first patched error has order -1. Repeating the same correction in the global symbol sequence removes successively all homogeneous errors. Borel summation then gives a left parametrix Q_L and, by the analogous right recursion, a right parametrix Q_R:

$$Q_L P = I - R_L, \qquad P Q_R = I - R_R, qquad R_L, R_R \in \Psi^{-\infty}.$$

The single operator

$$Q = Q_L + Q_R - Q_R P Q_L$$

is two-sided modulo smoothing: substituting the two identities shows that both $QP - I$ and $PQ - I$ contain a smoothing factor. This is the standard globalization and asymptotic-completeness argument of the classical pseudodifferential calculus. □

Let $H^s(M;E)$ denote the Sobolev space of order s. An operator in Ψ^m extends boundedly

$$P : H^s(M;E) \longrightarrow H^{s-m}(M;F).$$

A smoothing operator maps every Sobolev space continuously into every higher Sobolev space and is compact as an operator $H^s \to H^s$.

$$
\begin{tikzcd}
P \arrow[r, "\sigma_m"] & p_m \arrow[r, "\text{invert}"] & p_m^{-1} \arrow[d, "\text{recursive corrections}"] \\
\operatorname{ind} P & I - \text{compact} \arrow[l, "\text{Atkinson}"'] & Q \arrow[l, "\text{compose}"']
\end{tikzcd}
$$

Figure 2.1: Ellipticity is converted into Fredholmness through symbol inversion, quantization, and compact remainders.

Theorem 2.5 (Elliptic regularity)**.** *If P is elliptic, $u \in \mathcal{D}'(M;E)$, and $Pu \in H^{s-m}(M;F)$, then $u \in H^s(M;E)$. More precisely, for $u \in H^s$,*

$$\|u\|_{H^s} \le C\big(\|Pu\|_{H^{s-m}} + \|u\|_{H^{s-1}}\big).$$

Proof. Every distribution on a compact manifold belongs to H^{-N} for some N. The parametrix identity, valid on distributions,

$$u = QPu + R_L u,$$

places the first term in H^s and the second in C^∞, proving regularity directly. For the estimate, use the mapping properties $Q : H^{s-m} \to H^s$ and $R_L : H^{s-1} \to H^s$:

$$\|u\|_{H^s} \le \|QPu\|_{H^s} + \|R_L u\|_{H^s} \le C\big(\|Pu\|_{H^{s-m}} + \|u\|_{H^{s-1}}\big).$$

This also records the lower-order term needed when the estimate is used in a compactness argument. □

Corollary 2.6. *On a compact manifold, an elliptic operator*

$$P : H^s(M;E) \longrightarrow H^{s-m}(M;F)$$

is Fredholm. Its kernel consists of smooth sections, and

$$\operatorname{coker} P \cong \ker P^*.$$

Proof. The parametrix identities make P invertible modulo compact operators, so Atkinson's theorem gives Fredholmness. Elliptic regularity makes the kernel smooth. The cokernel is the orthogonal complement of the range, which is $\ker P^*$. □

Theorem 2.7 (Hörmander–Fedosov trace formula)**.** *Let $Q \in \Psi^{-m}(M;F,E)$ be any parametrix for P and put*

$$R_E = I - QP, \qquad R_F = I - PQ.$$

Then both remainders are smoothing and

$$\operatorname{ind} P = \operatorname{Tr}(R_E) - \operatorname{Tr}(R_F).$$

Proof. Let Π_E and Π_F be the orthogonal projections onto $\ker P$ and $\ker P^*$. They have finite rank and smooth kernels. The generalized inverse $G : (\ker P^*)^\perp \to (\ker P)^\perp$, extended by zero on $\ker P^*$, satisfies

$$GP = I - \Pi_E, \qquad PG = I - \Pi_F.$$

The parametrix construction, followed by these finite-rank corrections, shows that $G - Q$ is smoothing. Put $K = G - Q$. Then

$$R_E = \Pi_E + KP, \qquad R_F = \Pi_F + PK.$$

On a closed manifold, a smoothing operator is trace class and its composition with a differential or pseudodifferential operator is again smoothing. Kernel composition and Fubini's theorem give $\mathrm{Tr}(KP) = \mathrm{Tr}(PK)$. Therefore

$$\mathrm{Tr}(R_E) - \mathrm{Tr}(R_F) = \mathrm{Tr}(\Pi_E) - \mathrm{Tr}(\Pi_F) = \dim\ker P - \dim\ker P^*.$$

This is the analytic trace formula used implicitly in the original heat-kernel discussion. □

Question 2.8. *Where are compactness of M and ellipticity used in Fredholmness?*

Answer. Ellipticity produces a parametrix whose remainders gain arbitrarily many derivatives. Compactness of M makes the Sobolev inclusion $H^{s+N} \hookrightarrow H^s$ compact, so those remainders are compact. On a noncompact manifold, translates of one bump function can remain bounded with no convergent subsequence; smoothing alone then need not imply compactness. The boundary calculus in the companion notes is designed to control exactly that failure. □

The analytic index is

$$\mathrm{ind}\, P = \dim\ker P - \dim\ker P^*.$$

It is independent of s, invariant under continuous elliptic deformations, and unchanged by lower-order perturbations.

2.3 Hodge theory

The original lectures reached Hodge theory twice: first from an elliptic parametrix and later by integrating the heat parametrix. Both viewpoints are useful. The first proves closed range and finite-dimensionality; the second identifies the Green operator geometrically.

Let $(E^\bullet, D_\bullet)$ be an elliptic complex, so $D_j D_{j-1} = 0$, and put

$$\Delta_j = D_{j-1}D_{j-1}^* + D_j^* D_j, \qquad \mathcal{H}^j = \ker \Delta_j.$$

Lemma 2.9 (Energy identity)**.** *For every smooth section $u \in \Gamma(E^j)$,*

$$\langle \Delta_j u, u\rangle = \|D_{j-1}^* u\|^2 + \|D_j u\|^2.$$

Consequently

$$\mathcal{H}^j = \ker D_j \cap \ker D_{j-1}^*.$$

Proof. Move one factor in each summand of Δ_j across the inner product. Both terms are nonnegative, so their sum vanishes precisely when both operators annihilate u. □

The analytic input is the following estimate. It is the point at which ellipticity and compactness of the manifold enter the proof.

Proposition 2.10 (Elliptic estimate and compact resolvent)**.** *For every $s \in \mathbb{R}$ there is a constant C_s such that*

$$\|u\|_{H^{s+2}} \le C_s\big(\|\Delta_j u\|_{H^s} + \|u\|_{H^s}\big).$$

The closure of Δ_j initially defined on smooth sections is the self-adjoint operator with domain $H^2(E^j)$. It has compact resolvent. Hence its spectrum is a discrete sequence

$$0 \le \lambda_0 \le \lambda_1 \le \cdots \longrightarrow \infty$$

with finite multiplicities and a smooth orthonormal eigenbasis.

Proof. Choose a pseudodifferential parametrix Q of order -2 with $Q\Delta_j = I - R$, where R is smoothing. Mapping properties of Q and R give

$$\|u\|_{H^{s+2}} \le C\|\Delta_j u\|_{H^s} + C'\|u\|_{H^s}.$$

The symmetric nonnegative operator defined by the quadratic form

$$q_j(u) = \|D_j u\|^2 + \|D_{j-1}^* u\|^2, \qquad u \in H^1(E^j),$$

has a Friedrichs extension. The first-order elliptic estimate for $D_j \oplus D_{j-1}^*$ makes $q_j(u) + \|u\|^2$ equivalent to the H^1-norm; in particular the form is closed and its form domain really is $H^1(E^j)$. Elliptic regularity identifies its operator domain with $H^2(E^j)$, so this is the unique self-adjoint closure of the smooth Laplacian. For every $c > 0$, the energy identity makes $\Delta_j + c$ injective. Fredholmness and self-adjointness identify its cokernel with its kernel, hence it is bijective and its inverse maps L^2 boundedly to H^2. The inclusion $H^2 \hookrightarrow L^2$ is compact by Rellich's lemma, so $(\Delta_j + c)^{-1}$ is compact and self-adjoint on L^2. The spectral theorem for compact self-adjoint operators gives the asserted eigenbasis; elliptic regularity makes every eigenvector smooth. □

Theorem 2.11 (Hodge decomposition)**.** *For every j,*

$$L^2(E^j) = \mathcal{H}^j \mathbin{\widehat{\oplus}} \operatorname{im} D_{j-1} \mathbin{\widehat{\oplus}} \operatorname{im} D_j^*.$$

Here the operators are taken between their natural Sobolev domains; both ranges are closed. On smooth sections the decomposition restricts to

$$\Gamma(E^j) = \mathcal{H}^j \oplus D_{j-1}\Gamma(E^{j-1}) \oplus D_j^*\Gamma(E^{j+1}).$$

Moreover $\mathcal{H}^j$ is finite-dimensional and

$$H^j(D) \cong \mathcal{H}^j.$$

Proof. Let H_j be orthogonal projection onto the zero eigenspace. On the orthogonal complement define the Green operator by the convergent spectral series

$$G_j u = \sum_{\lambda_\nu > 0} \lambda_\nu^{-1}\langle u, e_\nu\rangle e_\nu, \qquad G_j H_j = H_j G_j = 0.$$

The first positive eigenvalue is separated from zero, so G_j is bounded on L^2, and

$$\Delta_j G_j = G_j \Delta_j = I - H_j.$$

The elliptic estimate applied repeatedly shows that G_j maps smooth sections to smooth sections.

On smooth sections the complex identities give

$$D_j\Delta_j = \Delta_{j+1}D_j, \qquad D^*_{j-1}\Delta_j = \Delta_{j-1}D^*_{j-1}.$$

Every eigenvector is smooth. Thus, if $\Delta_j e_\nu = \lambda_\nu e_\nu$, then $D_j e_\nu$ is either zero or a smooth λ_ν-eigenvector of Δ_{j+1}, and similarly for D^*_{j-1}. Applying the spectral definition of G on the finite linear span of the eigenvectors therefore gives

$$D_jG_j = G_{j+1}D_j, \qquad D^*_{j-1}G_j = G_{j-1}D^*_{j-1}.$$

These identities extend to their natural graph domains because that span is a core for all the operators involved. Moreover $G_j : L^2 \to H^2$ is bounded by the elliptic estimate.

Expanding $I - H_j = \Delta_j G_j$ now yields the explicit decomposition

$$u = H_j u + D_{j-1}D^*_{j-1}G_j u + D^*_j D_j G_j u.$$

All three displayed operators are bounded on L^2: the Green operator gains two derivatives and each remaining factor has total order two. The three terms are mutually orthogonal. For example,

$$\langle D_{j-1}v, D^*_j w\rangle = \langle D_j D_{j-1}v, w\rangle = 0.$$

Put

$$P_{\mathrm{ex}} = D_{j-1}D^*_{j-1}G_j, \qquad P_{\mathrm{coex}} = D^*_j D_j G_j.$$

The decomposition just proved is orthogonal, and $P_{\mathrm{ex}}u$ is exact while $P_{\mathrm{coex}}u$ is coexact. Applying the decomposition to either of these two terms shows

$$P^2_{\mathrm{ex}} = P_{\mathrm{ex}}, \qquad P^2_{\mathrm{coex}} = P_{\mathrm{coex}}, \qquad P_{\mathrm{ex}}P_{\mathrm{coex}} = 0.$$

Their ranges are exactly the Sobolev ranges in the theorem. Indeed, $P_{\mathrm{ex}}u = D_{j-1}(D^*_{j-1}G_j u)$ lies in $D_{j-1}H^1$; conversely, for $v = D_{j-1}w$, first with w smooth and then by H^1-approximation, orthogonality and $I = H_j + P_{\mathrm{ex}} + P_{\mathrm{coex}}$ give $P_{\mathrm{ex}}v = v$. The coexact case is identical. Ranges of bounded projections are closed. Compact resolvent makes $\mathcal{H}^j$ finite-dimensional, and elliptic regularity makes it smooth.

If $D_j u = 0$, pair u with its coexact summand:

$$\langle u, D^*_j w\rangle = \langle D_j u, w\rangle = 0.$$

Orthogonality therefore forces that summand to vanish, so u is cohomologous to $H_j u$. A harmonic exact form is zero, proving uniqueness. Finally, if u is smooth, then $G_j u$ is smooth by elliptic regularity, so every term in the decomposition is smooth. This proves the smooth statement as well as the L^2 statement. □

2.3.1 The heat-kernel proof and the Green operator

The second proof in the original lectures starts from the heat operator. For $t > 0$, spectral calculus gives

$$e^{-t\Delta_j}u = \sum_\nu e^{-t\lambda_\nu}\langle u, e_\nu\rangle e_\nu.$$

It is smoothing, solves

$$(\partial_t + \Delta_j)e^{-t\Delta_j} = 0, \qquad \lim_{t\downarrow 0} e^{-t\Delta_j}u = u,$$

and converges in operator norm to H_j as $t \to \infty$. Therefore

$$G_j = \int_0^\infty (e^{-t\Delta_j} - H_j)\, dt.$$

The integral converges in operator norm because the nonzero spectrum is bounded below by the first positive eigenvalue. Integrating the heat equation gives

$$\Delta_j G_j = -\int_0^\infty \partial_t (e^{-t\Delta_j} - H_j)\, dt = I - H_j.$$

Thus the heat-parametrix construction in the next chapter does more than produce a fundamental solution: after integration in time it produces the same Green operator used in Hodge decomposition.

Question 2.12. *Does the heat-kernel proof assume Hodge decomposition when it uses the projection H_j?*

Answer. No. Compact resolvent is obtained from the elliptic estimate and Rellich compactness, independently of Hodge decomposition. The spectral theorem then defines the finite-dimensional zero-eigenspace and its orthogonal projection. Integrating the heat semigroup constructs G_j, and the identity $I - H_j = \Delta_j G_j$ yields Hodge decomposition. One may instead construct a heat parametrix first and use analytic Fredholm theory; the logical input is the same compactness statement. □

For the de Rham complex,

$$d^* = (-1)^{n(k+1)+1} * d * \quad \text{on } \Omega^k(M), \qquad \Delta = dd^* + d^*d.$$

Thus every de Rham cohomology class has a unique harmonic representative.

Corollary 2.13 (Poincaré duality). *If M is closed and oriented of dimension n, then the pairing*

$$H^k_{\mathrm{dR}}(M) \times H^{n-k}_{\mathrm{dR}}(M) \longrightarrow \mathbb{R}, \qquad ([\alpha],[\beta]) \longmapsto \int_M \alpha \wedge \beta$$

is nondegenerate.

Proof. Choose the harmonic representative α of a nonzero class. The Hodge star commutes with the Laplacian, so $*\alpha$ is harmonic and represents a class in degree $n-k$. Then

$$\int_M \alpha \wedge *\alpha = \|\alpha\|^2_{L^2} > 0.$$

Hence no nonzero class lies in the left radical; symmetry gives the same conclusion for the right radical. □

Grade forms by parity and set

$$D_{\mathrm{dR}} = d + d^* : \Omega^{\mathrm{ev}}(M) \longrightarrow \Omega^{\mathrm{odd}}(M).$$

Hodge theory gives

$$\operatorname{ind} D_{\mathrm{dR}} = \chi(M).$$

2.4 The symbol class and the index theorem

If $P : E \to F$ is elliptic, its principal symbol is an isomorphism over $T^*M \setminus 0$. Hence the triple

$$(\pi^* E, \pi^* F, \sigma(P))$$

defines a compactly supported class

$$[\sigma(P)] \in K_c^0(T^*M).$$

Homotopic elliptic symbols determine the same class.

Let $\mathcal{E}(M)$ denote stable homotopy classes of elliptic operators. The two notions of index fit into the diagram

$$\begin{tikzcd}
\mathcal{E}(M) \arrow[r, "{[\sigma]}"] \arrow[dr, "\mathrm{ind}"'] & K_c^0(T^*M) \arrow[d, "\text{t}-\mathrm{ind}"] \\
& \mathbb{Z}.
\end{tikzcd}$$

Figure 2.2: The Atiyah–Singer theorem states that the analytic index factors through the compactly supported symbol class.

Theorem 2.14 (Atiyah–Singer index theorem)**.** *There is a topological index homomorphism*

$$\text{t-ind} : K_c^0(T^*M) \longrightarrow \mathbb{Z}$$

constructed from the Thom isomorphism and Bott periodicity, and every elliptic operator satisfies

$$\operatorname{ind} P = \text{t-ind}[\sigma(P)].$$

After applying the compactly supported Chern character, the topological index becomes the characteristic-number pairing determined by the Thom class and $\mathrm{Td}(T_{\mathbb{C}}M)$*.*

The theorem says that the analytic index depends only on the homotopy class of the principal symbol. For Dirac operators, the Thom-class calculation reduces the general characteristic-number formula to the explicit $\widehat{A}$-formula proved by heat kernels below.

To make the construction of t-ind concrete, embed M in an even-dimensional Euclidean space $\mathbb{R}^N$. Choose metrics to identify tangent and cotangent bundles. The differential of the embedding then gives $T^*M \hookrightarrow T^*\mathbb{R}^N$. If ν is the normal bundle of M, the normal bundle of this induced embedding is $\pi^*(\nu \oplus \nu)$; the rule $J(v, w) = (-w, v)$ gives it a canonical complex structure. Its Thom class defines a push-forward

$$K_c^0(T^*M) \longrightarrow K_c^0(T^*\mathbb{R}^N).$$

Bott periodicity identifies the target with $\mathbb{Z}$; the resulting integer is the topological index. Homotopy invariance of the Thom class and stabilization show that it is independent of the embedding.

Discussion. The general Atiyah–Singer theorem above uses two major topological inputs: the Thom isomorphism and Bott periodicity. The heat-kernel argument in the following chapters proves the local formula for Dirac operators. The standard K-theoretic reduction of a general elliptic symbol to Dirac symbols then recovers the general theorem. Keeping these two layers separate prevents the Dirac calculation from being mistaken for a proof that silently assumes every elliptic operator is already of Dirac type.

2.5 The K-theoretic proof

The preceding construction defines the topological index. We now explain why it equals the analytic index. The proof is short only after three substantial facts have been isolated: an elliptic symbol can be quantized, the analytic index commutes with Thom push-forward, and the Bott generator has index one. This is the standard K-theoretic proof architecture of Atiyah and Singer [1, 2].

First define the analytic index directly on compactly supported K-theory. A class in $K^0_c(T^*M)$ is represented, after stabilization, by

$$\alpha = [\pi^*E, \pi^*F, a],$$

where $a : \pi^*E \to \pi^*F$ is invertible outside a compact set. Radial deformation makes a homogeneous for large covectors, and symbol quantization produces an elliptic pseudodifferential operator $P_a : E \to F$ with principal symbol a. Set

$$\text{a-ind}(\alpha) = \operatorname{ind} P_a.$$

This integer is well defined. A homotopy of symbols quantizes to a continuous family of Fredholm operators between fixed Sobolev spaces, so its index is constant. Direct sum gives additivity, and adding the symbol of an invertible bundle map adds an invertible operator and hence index zero. These are precisely the relations used to define $K^0_c(T^*M)$.

Theorem 2.15 (Analytic push-forward; Atiyah–Singer)**.** *Let $i : M \hookrightarrow N$ be an embedding of closed manifolds. For every $\alpha \in K^0_c(T^*M)$,*

$$\text{a-ind}(i_!\alpha) = \text{a-ind}(\alpha).$$

Proof architecture and imported inputs. Choose a tubular neighborhood with normal bundle ν. The normal bundle of T^*M in T^*N is $\pi^*(\nu \oplus \nu)$, with complex structure

$$J(v, w) = (-w, v).$$

Consequently $i_!$ is, in the tubular neighborhood, exterior product with the Bott–Thom class, followed by extension to T^*N.

Here is the analytic representative of that class. On a Euclidean fibre V act on $\Lambda^\bullet V^*_{\mathbb{C}}$ by

$$c(v) = \varepsilon(v^\flat) - \iota_v, \qquad \widehat{c}(v) = \varepsilon(v^\flat) + \iota_v.$$

For an orthonormal basis $e_1, \ldots, e_r$, the odd Bott operator is

$$B_V = \sum_{j=1}^{r} c(e_j)\frac{\partial}{\partial v_j} + \widehat{c}(v).$$

Its full symbol in the position and covariable pair (v, ξ) is $i\, c(\xi) + \widehat{c}(v)$, the Bott class on $V \oplus V^*$. With the conventions above,

$$B_V^2 = -\Delta + |v|^2 + \sum_{j=1}^{r} c(e_j)\widehat{c}(e_j).$$

The harmonic-oscillator decomposition shows that its kernel is the one-dimensional Gaussian vacuum in even Bott degree and that its odd kernel is zero. In particular, B_V is Fredholm and

$$\operatorname{ind} B_V = 1.$$

Write P_a as the odd self-adjoint operator

$$\mathcal{P}_a = \begin{pmatrix} 0 & P_a^* \\ P_a & 0 \end{pmatrix} \quad \text{on } E \oplus F.$$

Let $\Gamma_{E\oplus F}$ act by $+1$ on E and by -1 on F. Over the tubular neighborhood, the graded product

$$\mathcal{P}_a \widehat{\otimes} 1 + \Gamma_{E\oplus F} \widehat{\otimes} B_\nu$$

has symbol α multiplied by the Thom class. Its kernel is the kernel of $\mathcal{P}_a$ tensored with the Gaussian vacuum; equivalently, the Fredholm product formula gives

$$\operatorname{ind}(P_a \,\#\, B_\nu) = \operatorname{ind}(P_a) \operatorname{ind}(B_\nu) = \operatorname{ind}(P_a).$$

A complete proof uses three standard results, each in its family version: the Bott operator above represents the Thom class and has index one; the analytic index is multiplicative for the graded external product; and quantization is compatible with tubular-neighborhood excision. A connection globalizes the fibrewise Bott operator. In the exact product model the two odd summands anticommute, so the square splits into base and normal parts. For a general connection, differentiating the vertical Bott operator produces lower-order mixed terms; the usual rescaling argument shows that they do not change the Fredholm class. Changing the connection or tubular neighborhood likewise produces a Fredholm homotopy. Excision then extends the tubular model to N, since the symbol is invertible on the complement. These are the analytic push-forward theorem of Atiyah–Singer and the Atiyah–Bott–Shapiro Thom construction; full proofs are given in [1, 2]. Subject to these three quoted inputs, the displayed product-index calculation proves the claim. □

We can now finish the proof of the index theorem. Choose an embedding $i : M \hookrightarrow \mathbb{R}^N$. Bott periodicity gives

$$K_c^0(T^*\mathbb{R}^N) \cong K_c^0(\mathbb{R}^{2N}) \cong \mathbb{Z}.$$

Let β_N be the Bott generator, normalized by $\text{a-ind}(\beta_N) = 1$. If $i_!\alpha = k\beta_N$, then the push-forward lemma gives

$$\text{a-ind}(\alpha) = \text{a-ind}(i_!\alpha) = k.$$

By definition the topological index of α is the same integer k. Therefore

$$\text{a-ind} = \text{t-ind} \quad \text{on } K_c^0(T^*M),$$

which proves the Atiyah–Singer theorem for every elliptic operator. Applying the compactly supported Chern character and the Thom identity turns this equality into the cohomological formula. For a Dirac symbol, the characteristic classes simplify to the $\widehat{A}$ ch-density computed by the heat kernel.

Question 2.16. *Where is the difficult geometry hidden in this short proof?*

Answer. It is concentrated in the push-forward lemma. Quantization converts symbol homotopy into Fredholm homotopy; the Thom class becomes the fibrewise Bott operator; and excision identifies a tubular neighborhood with its normal model. Once those compatibilities and $\operatorname{ind} B_V = 1$ are proved, the final reduction to $K_c^0(\mathbb{R}^{2N}) \cong \mathbb{Z}$ is formal. □

Chapter 3

Heat kernels

3.1 Existence and basic properties

Let L be a nonnegative, formally self-adjoint elliptic differential operator of order two on a Hermitian bundle $E \to M$. Its self-adjoint extension has discrete spectrum

$$0 \leq \lambda_0 \leq \lambda_1 \leq \cdots, \qquad \lambda_j \longrightarrow \infty,$$

with a complete orthonormal basis of smooth eigensections $\{\phi_j\}$.

For $t > 0$, functional calculus defines

$$e^{-tL}u = \sum_j e^{-t\lambda_j} \langle u, \phi_j \rangle \phi_j.$$

Its Schwartz kernel is

$$K(t; x, y) = \sum_j e^{-t\lambda_j} \phi_j(x) \otimes \phi_j(y)^*.$$

The series converges in every smooth topology for $t \geq t_0 > 0$. Consequently e^{-tL} is smoothing and trace class, and

$$\mathrm{Tr}(e^{-tL}) = \sum_j e^{-t\lambda_j}.$$

The heat operator satisfies

$$(\partial_t + L)K = 0, \qquad \lim_{t \downarrow 0} e^{-tL}u = u,$$

and the semigroup identity

$$e^{-(t+s)L} = e^{-tL}e^{-sL}.$$

Proposition 3.1 (Duhamel formula)**.** *Let L_r be a differentiable family of nonnegative self-adjoint Laplace-type operators with common domain $H^2(E)$, and suppose $r \mapsto L_r$ is differentiable as a map $H^2(E) \to L^2(E)$. Then*

$$\frac{d}{dr} e^{-tL_r} = -\int_0^t e^{-(t-s)L_r} \dot{L}_r e^{-sL_r} \, ds.$$

Proof. Let $U_r(t)$ denote the right side. Differentiation of $(\partial_t + L_r)e^{-tL_r} = 0$ with respect to r shows that $\partial_r e^{-tL_r}$ solves

$$(\partial_t + L_r)U = -\dot{L}_r e^{-tL_r}, \qquad U(0) = 0.$$

Variation of constants gives the displayed integral. Uniqueness of the inhomogeneous heat equation completes the proof. $\square$

3.2 The local parametrix

An operator is of Laplace type if its principal symbol is

$$\sigma_2(L)(x,\xi) = |\xi|^2 \,\mathrm{Id}_{E_x}.$$

Every Laplace-type operator has a unique expression

$$L = \nabla^*\nabla + \mathcal{R}$$

for a connection ∇ and an endomorphism $\mathcal{R}$.

Near the diagonal, let $r(x,y)$ be the Riemannian distance. Seek a parametrix of the form

$$H_N(t;x,y) = (4\pi t)^{-n/2} e^{-r(x,y)^2/(4t)} \sum_{j=0}^{N} t^j u_j(x,y).$$

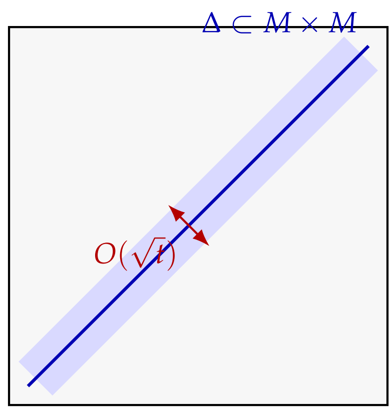

Figure 3.1: As $t \downarrow 0$, the heat kernel concentrates in a $\sqrt{t}$-neighborhood of the diagonal; the transport equations resolve its coefficients along radial geodesics.

Substitution into $(\partial_t + L_x)H_N = 0$ first determines u_0 by parallel transport and the Riemannian density, then determines each u_j from a transport equation along the radial geodesic.

More explicitly, let $\gamma(\rho) = \exp_y(\rho v)$, $0 \le \rho \le 1$, with $x = \exp_y v$. If $J(x,y)$ is the Jacobian of the exponential map and $\tau(x,y) : E_y \to E_x$ is parallel transport, then

$$u_0(x,y) = J(x,y)^{-1/2}\tau(x,y), \qquad u_0(y,y) = I.$$

After conjugation by u_0, the j-th transport equation is an ordinary radial equation. Its solution is

$$u_j(x,y) = -u_0(x,y)\int_0^1 \rho^{j-1} u_0(\gamma(\rho),y)^{-1}(L_x u_{j-1})(\gamma(\rho),y)\,d\rho, \qquad j \ge 1.$$

The integral is smooth at $x = y$, and induction gives smooth u_j on a normal neighborhood of the diagonal.

Multiply by a cutoff supported in a normal neighborhood of the diagonal. For every pair of nonnegative integers a, b, the error

$$R_N = (\partial_t + L)H_N$$

satisfies, after solving sufficiently many transport equations,

$$\|\partial_t^a R_N(t)\|_{C^b(M\times M)} \le C_{a,b,N} t^{N-n/2-a-b/2}.$$

Each spatial derivative can cost a factor $t^{-1/2}$; the stronger estimate without $b/2$ would be false for a fixed truncation. For N large relative to the requested seminorms, the Volterra series

$$K = H_N - H_N * R_N + H_N * R_N * R_N - \cdots$$

converges in the Fréchet topology of smooth kernels on $(0, T] \times M \times M$, with its natural weighted seminorms at $t = 0$, where

$$(A * B)(t) = \int_0^t A(t-s)B(s)\, ds.$$

It solves the heat equation with the correct initial value and therefore equals the heat kernel.

The algebraic cancellation uses the distributional identity $H_N(0) = I$. For any residual kernel S, differentiation of the upper convolution limit gives

$$(\partial_t + L)(H_N * S) = S + R_N * S.$$

Consequently successive terms in the Volterra series cancel pairwise. This boundary term is essential; without it the series would be only a formal Neumann expression.

For completeness, the Volterra series converges because its k-fold term is integrated over an ordered simplex. If $\|R_N(t)\| \le Ct^\alpha$, $\alpha > -1$, then repeated convolution gives

$$\|R_N^{*k}(t)\| \le \frac{C^k \Gamma(\alpha+1)^k}{\Gamma(k(\alpha+1))} t^{k(\alpha+1)-1}.$$

The gamma function supplies factorial decay. The same estimate for derivatives gives convergence in every smooth kernel seminorm.

Theorem 3.2 (Heat-kernel expansion)**.** *On the diagonal,*

$$K(t; x, x) \sim (4\pi t)^{-n/2} \sum_{j=0}^{\infty} t^j a_j(x) \qquad (t \downarrow 0),$$

in the smooth topology. Each $a_j(x)$ is a universal polynomial in the curvatures of M and E, the endomorphism $\mathcal{R}$, and their covariant derivatives. In particular,

$$a_0 = \mathrm{Id}_E, \qquad a_1 = \frac{1}{6}\,\mathrm{scal}\ \mathrm{Id}_E - \mathcal{R}$$

for the convention $L = \nabla^*\nabla + \mathcal{R}$.

Integrating the fibrewise trace gives

$$\mathrm{Tr}(e^{-tL}) \sim (4\pi t)^{-n/2} \sum_{j=0}^{\infty} t^j \int_M \mathrm{tr}_E\, a_j(x)\, d\,\mathrm{vol}(x).$$

3.3 McKean–Singer

Let

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}$$

be an odd, formally self-adjoint elliptic operator on a graded bundle $E^+ \oplus E^-$. Then

$$D^2 = \begin{pmatrix} D^-D^+ & 0 \\ 0 & D^+D^- \end{pmatrix}.$$

For a trace-class endomorphism T, define

$$\operatorname{Str}(T) = \operatorname{Tr}(T|_{E^+}) - \operatorname{Tr}(T|_{E^-}).$$

Theorem 3.3 (McKean–Singer)**.** *For every $t > 0$,*

$$\operatorname{Str}(e^{-tD^2}) = \operatorname{ind} D^+.$$

Proof. Every positive eigenvalue λ of D^-D^+ is paired with the same eigenvalue of D^+D^-: the map D^+ is an isomorphism between the corresponding eigenspaces, with inverse $\lambda^{-1}D^-$. Their heat traces cancel in the supertrace. The zero eigenspaces contribute $\dim\ker D^+ - \dim\ker D^-$, which is the index. □

Question 3.4. *Why do the zero modes survive while every positive mode cancels?*

Answer. On a positive eigenspace, D^+ has inverse $\lambda^{-1}D^-$, so the even and odd multiplicities are equal and carry the same heat weight $e^{-t\lambda}$. At eigenvalue zero this inverse does not exist. The unpaired spaces are precisely $\ker D^+$ and $\ker D^-$, whose signed dimension is the index. □

Equivalently, cyclicity of the supertrace gives

$$\frac{d}{dt}\operatorname{Str}(e^{-tD^2}) = -\frac{1}{2}\operatorname{Str}([D, De^{-tD^2}]_{\mathrm{s}}) = 0.$$

The small-time limit is therefore the index.

When the pointwise limit exists and

$$\operatorname{str} K(t; x, x) \sim (4\pi t)^{-n/2} \sum_{j\geq 0} t^j \operatorname{str} a_j(x),$$

then

$$\operatorname{ind} D^+ = \int_M \lim_{t\downarrow 0} \operatorname{str} K(t; x, x)\, d\operatorname{vol}(x).$$

For a general elliptic operator, this formula still computes the index after replacing it by the odd self-adjoint operator

$$\mathcal{D} = \begin{pmatrix} 0 & P^* \\ P & 0 \end{pmatrix}.$$

The remaining task is to identify the constant term of the local supertrace. For Dirac operators, Clifford symmetry forces the singular terms to cancel and Getzler rescaling computes the survivor.

Chapter 4

Clifford modules and Dirac operators

4.1 Clifford algebras

For a Euclidean vector space V, the complex Clifford algebra is

$$\mathrm{Cl}(V) = T(V_{\mathbb{C}})/\langle vw + wv + 2\langle v, w\rangle\rangle.$$

Thus

$$c(v)c(w) + c(w)c(v) = -2\langle v, w\rangle.$$

The algebra is $\mathbb{Z}_2$-graded by tensor parity.

If $\dim V = 2m$, the complex spin representation S has dimension 2^m and a grading

$$S = S^+ \oplus S^-.$$

Clifford multiplication by a vector is odd. The grading operator is the complex volume element, normalized to square to one and to agree with the chosen orientation.

Definition 4.1. A graded Clifford module over an oriented even-dimensional Riemannian manifold is a graded Hermitian bundle

$$E = E^+ \oplus E^-$$

with a fibrewise action

$$c : \mathrm{Cl}(T^*M) \longrightarrow \mathrm{End}(E)$$

such that $c(\xi)$ is odd and skew-adjoint for real ξ.

Locally, every complex Clifford module in even dimension is of the form

$$E \cong S \otimes W,$$

where the Clifford algebra acts on S and trivially on the twisting factor W. The bundle W need not exist globally unless a spinor bundle does, but its Chern character is globally represented by the relative curvature introduced below.

4.2 Clifford connections

Definition 4.2. A connection ∇^E on a Clifford module is a Clifford connection if

$$[\nabla^E_X, c(\alpha)] = c(\nabla^{T^*M}_X \alpha)$$

for every vector field X and one-form α.

The associated Dirac operator is

$$D = c \circ \nabla^E = \sum_{j=1}^n c(e^j)\nabla^E_{e_j}$$

in a local orthonormal frame. It is odd and formally self-adjoint. Its principal symbol is

$$\sigma_1(D)(x,\xi) = i\, c(\xi).$$

Since

$$(i\, c(\xi))^2 = |\xi|^2 \,\mathrm{Id},$$

the operator is elliptic.

The curvature of a Clifford connection splits into its Clifford and twisting parts. Define

$$F^{E/S}(X,Y) = F^E(X,Y) - \frac{1}{4}\sum_{j,k}\langle R^{TM}(X,Y)e_j, e_k\rangle c(e^j)c(e^k).$$

This endomorphism commutes with Clifford multiplication. Locally, for $E = S \otimes W$, it is exactly F^W.

For an endomorphism-valued two-form F, write

$$c(F) = \frac{1}{2}\sum_{j,k} c(e^j)c(e^k)F(e_j,e_k).$$

Theorem 4.3 (Weitzenböck–Lichnerowicz formula)**.** *A Dirac operator satisfies*

$$D^2 = (\nabla^E)^*\nabla^E + \frac{1}{4}\,\mathrm{scal} + c(F^{E/S}).$$

Proof. At a chosen point, take a synchronous orthonormal frame, so $\nabla e_j = 0$ there. The symmetric part of $\sum_{j,k} c(e^j)c(e^k)\nabla_{e_j}\nabla_{e_k}$ is the connection Laplacian. The antisymmetric part is

$$\frac{1}{2}\sum_{j,k} c(e^j)c(e^k)F^E(e_j,e_k).$$

Insert the decomposition of F^E. The first Bianchi identity and the Clifford relations contract the Riemannian curvature term to $\mathrm{scal}/4$; the remainder is $c(F^{E/S})$.

For the contraction, write the spin-curvature contribution as

$$\frac{1}{8}\sum_{i,j,k,\ell} R_{ijk\ell}\, c(e^i)c(e^j)c(e^k)c(e^\ell).$$

Terms with four distinct indices cancel by the first Bianchi identity. Terms with exactly three distinct indices cancel by the skew symmetries of R. The remaining pairings reduce, using $c(e^i)^2 = -1$, to

$$\frac{1}{4}\sum_{i,j} R_{ijji} = \frac{1}{4}\,\mathrm{scal}.$$

$\square$

4.3 Basic examples

On differential forms, define
$$c(\xi) = \varepsilon(\xi) - \iota_{\xi^\sharp}.$$
The exterior and contraction operators satisfy
$$\varepsilon(\xi)\iota_{\eta^\sharp} + \iota_{\eta^\sharp}\varepsilon(\xi) = \langle \xi, \eta \rangle,$$
so c obeys the Clifford relation. The Levi–Civita connection on forms is a Clifford connection, and its Dirac operator is
$$D_{\mathrm{dR}} = d + d^*.$$
Its grading by even and odd degree gives
$$\operatorname{ind} D^+_{\mathrm{dR}} = \chi(M).$$

Example 4.4 (The Cauchy–Riemann operator as a Dirac operator)**.** On $\mathbb{R}^2 \cong \mathbb{C}$, with $z = x + iy$, set
$$\bar\partial = \frac{1}{2}(\partial_x + i\partial_y), \qquad \bar\partial^* = -\frac{1}{2}(\partial_x - i\partial_y).$$
Using the orthonormal frame $d\bar z/\sqrt{2}$ for $(0, 1)$-forms, the Dolbeault–Dirac operator has the scalar matrix
$$\mathcal{D}_{\bar\partial} = 2\begin{pmatrix} 0 & \bar\partial^* \\ \bar\partial & 0 \end{pmatrix}$$
and principal symbol
$$\sigma_1(\mathcal{D}_{\bar\partial})(\xi) = i\begin{pmatrix} 0 & -(\xi_x - i\xi_y) \\ \xi_x + i\xi_y & 0 \end{pmatrix},$$
whose square is $|\xi|^2$ times the identity. This is the two-dimensional spin^c Dirac symbol and explains why the Dolbeault operator later has the same elliptic symbol as a Dirac operator.

If M is spin, its spinor bundle $S = S^+ \oplus S^-$ carries the spin connection and spin Dirac operator
$$D_S : \Gamma(S^+) \longrightarrow \Gamma(S^-).$$
For a Hermitian bundle W with unitary connection, the twisted operator on $S \otimes W$ satisfies
$$D_W^2 = \nabla^*\nabla + \frac{1}{4}\operatorname{scal} + c(F^W).$$

Corollary 4.5 (Lichnerowicz vanishing)**.** *If M is closed, spin, and has strictly positive scalar curvature, then the untwisted spin Dirac operator has zero kernel and zero index.*

Proof. For $D_S\psi = 0$, integration of the Lichnerowicz formula gives
$$0 = \|\nabla\psi\|^2 + \frac{1}{4}\int_M \operatorname{scal}|\psi|^2.$$
Strict positivity forces $\psi = 0$. □

4.4 Clifford supertrace

The supertrace on a graded endomorphism algebra is

$$\operatorname{str}(T) = \operatorname{tr}(T|_{E^+}) - \operatorname{tr}(T|_{E^-}).$$

It vanishes on supercommutators.

Lemma 4.6 (Clifford cancellation)**.** *Let* $\dim V = n$ *be even and let* S *be the spin representation. The supertrace of an element of reduced Clifford degree less than n is zero. Equivalently, the supertrace vanishes below degree n in the associated graded Clifford algebra. On the top Clifford degree, it is the oriented volume functional, with normalization fixed by the chirality operator.*

Proof. Use the Clifford relations first to reduce every word to a linear combination of products of distinct generators. Odd reduced monomials are odd endomorphisms and therefore have zero supertrace. Let $T = c(e^{i_1}) \cdots c(e^{i_{2r}})$ have distinct indices and even degree $2r < n$, and choose e^j not among its factors. Then $c(e^j)$ commutes with T but anticommutes with the chirality operator Γ. Since $\operatorname{str}(T) = \operatorname{tr}(\Gamma T)$, ordinary trace invariance under conjugation by $c(e^j)$ gives

$$\operatorname{str}(T) = \operatorname{tr}\Big(c(e^j)^{-1}\Gamma T c(e^j)\Big) = -\operatorname{str}(T),$$

so it vanishes. In degree n, the monomial is a scalar multiple of the complex volume element and hence of Γ; its supertrace is nonzero. The chosen orientation and the relation $c(e^j)^2 = -1$ fix that scalar. □

Discussion. The cancellation is pointwise. It removes the negative powers of t in the local heat supertrace before integration. McKean–Singer says the integral is independent of t; Clifford cancellation explains why its small-time limit is already a smooth top-degree differential form.

This cancellation is the algebraic reason the divergent coefficients in the Dirac heat supertrace vanish.

Chapter 5

The local index theorem

5.1 Relative curvature

For a graded Clifford module E, define the relative Chern character form by

$$\mathrm{ch}(E/S, \nabla^E) = \mathrm{tr}_{E/S} \exp\left(\frac{iF^{E/S}}{2\pi}\right).$$

The relative trace is characterized locally by

$$E \cong S \otimes W \quad \Longrightarrow \quad \mathrm{ch}(E/S) = \mathrm{ch}(W).$$

This prescription is independent of the local spinor factorization and therefore defines a global differential form. More explicitly, the endomorphisms commuting with the Clifford action form the commutant bundle $\mathrm{End}_{\mathrm{Cl}}(E)$. On a local factorization $E \cong S \otimes W$, this bundle is $I_S \otimes \mathrm{End}(W)$, and $\mathrm{tr}_{E/S}(I_S \otimes T) = \mathrm{tr}_W(T)$. Changes of spinor factorization act by conjugation on the commutant, so the trace patches. Since $F^{E/S}$ lies in this commutant and satisfies its own Bianchi identity, Chern–Weil transgression proves that the relative Chern character is closed and connection-independent in cohomology.

The local index density is

$$\left[\widehat{A}(TM, \nabla^{TM})\, \mathrm{ch}(E/S, \nabla^E)\right]_{[n]}.$$

5.2 Getzler rescaling

Fix $p \in M$. Choose geodesic normal coordinates centered at p and trivialize E by radial parallel transport. In these coordinates, the connection and metric have expansions whose first nonconstant terms are determined by curvature.

In radial gauge,

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3}R_{ikj\ell}(p)x^k x^\ell + O(|x|^3), \qquad \nabla_i^E = \partial_i - \frac{1}{2}F_{ij}^E(p)x^j + O(|x|^2).$$

The second formula follows by differentiating the radial-gauge condition $x^i\nabla_i^E = x^i\partial_i$: the linear part of the connection one-form is forced by its curvature. Splitting F^E into spin and relative curvature separates the part that acts through Clifford multiplication from the part commuting with it.

Assign Getzler degrees

$$\deg \partial_j = 1, \qquad \deg c(dx^j) = 1, \qquad \deg x^j = -1, \qquad \deg \partial_t = 2.$$

Under the parabolic scaling

$$x \mapsto \sqrt{t}\, x,$$

only the highest Getzler-order part of $\partial_t + D^2$ contributes to the constant term of the Clifford supertrace. The Clifford filtration is identified at its associated graded level with the exterior algebra:

$$c(dx^{i_1}) \cdots c(dx^{i_k}) \longmapsto dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

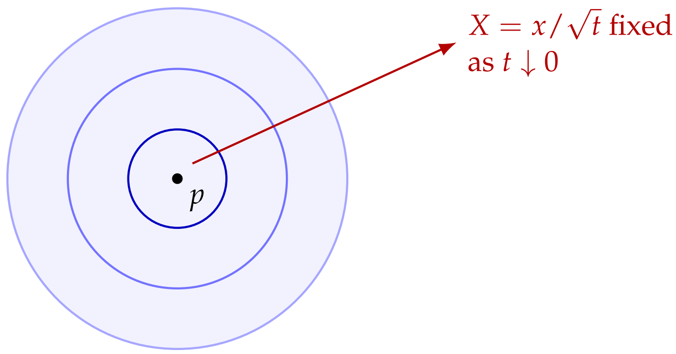

Figure 5.1: Getzler rescaling magnifies a shrinking neighborhood of p. The metric becomes Euclidean while the first nontrivial connection terms retain the curvature.

The rescaled model of D^2 is a harmonic oscillator with coefficients given by R_p^{TM}, together with the constant twisting curvature $F_p^{E/S}$. All terms of lower Getzler order disappear in the limit.

To state the model, put

$$\mathcal{R}_{ij} = \frac{1}{2} \sum_{k,\ell} R_{ijk\ell}(p)\, dx^k \wedge dx^\ell.$$

After passing from the Clifford filtration to its exterior associated graded, the model operator is

$$\mathcal{H}_p = -\sum_i \left(\partial_i - \frac{1}{4} \sum_j \mathcal{R}_{ij} x^j \right)^2 + F_p^{E/S}.$$

Here the curvature two-forms act by exterior multiplication in the associated graded Clifford algebra, while $F_p^{E/S}$ acts on the Clifford commutant. It is useful to separate the raw curvature in this differential operator from the normalized Chern–Weil variables

$$\Omega^{TM} = \frac{R^{TM}}{2\pi i}, \qquad \Omega^{E/S} = \frac{F^{E/S}}{2\pi i}.$$

With the convention of this manuscript,

$$\widehat{A}(TM) = \det{}^{1/2} \left(\frac{\Omega^{TM}/2}{\sinh(\Omega^{TM}/2)} \right), \qquad \mathrm{ch}(E/S) = \mathrm{tr}_{E/S}\, e^{-\Omega^{E/S}},$$

because $iF/(2\pi) = -F/(2\pi i)$. The scalar-curvature term in the Lichnerowicz formula has lower Getzler degree and disappears; its effect is already encoded in the oscillator through R_p^{TM}. If $\delta_\varepsilon u(x) = u(\varepsilon x)$ and Clifford degree k is multiplied by ε^{-k}, then

$$\varepsilon^2 \delta_\varepsilon^{-1} D^2 \delta_\varepsilon = \mathcal{H}_p + O(\varepsilon)$$

coefficientwise on compact sets. Duhamel's formula and Gaussian estimates then show convergence of the rescaled heat kernels, with all derivatives.

Lemma 5.1 (Model heat kernel; standard Mehler calculation)**.** *The heat kernel of the rescaled harmonic oscillator is given by Mehler's formula. After Clifford supertrace and the Chern–Weil normalization, its diagonal value at the origin is*

$$\left[\widehat{A}(TM, \nabla^{TM})\,\mathrm{ch}(E/S, \nabla^{E})\right]_{[n]}(p).$$

Normalization check. Put the skew-symmetric curvature matrix into 2×2 blocks. The model operator splits into commuting two-dimensional oscillators and a constant endomorphism. For the oscillator part, Mehler's formula at the origin is

$$K_{\mathrm{osc}}(t;0,0) = (4\pi t)^{-n/2} \det{}^{1/2}\left(\frac{t\mathcal{R}/2}{\sinh(t\mathcal{R}/2)}\right).$$

This identity is an equality of formal exterior-power series; it can be verified block by block from the one-dimensional Gaussian oscillator. Because $F_p^{E/S}$ commutes with the Clifford action, it commutes with the oscillator and contributes the raw factor $e^{-tF_p^{E/S}}$.

The symbol-supertrace normalization is the remaining step. Blockwise Fourier inversion contributes the powers of 2π, and the top Clifford supertrace changes the raw curvature variables to the normalized variables above. The precise identity is

$$\sigma_{[n]}\,\mathrm{str}\,K_{\mathcal{H}_p}(1;0,0) = \left[\det{}^{1/2}\left(\frac{\Omega^{TM}/2}{\sinh(\Omega^{TM}/2)}\right)\mathrm{tr}_{E/S}\,e^{-\Omega^{E/S}}\right]_{[n]}.$$

Thus multiplying the curvature blocks and taking degree n gives

$$\left[\widehat{A}(TM, \nabla^{TM})\,\mathrm{ch}(E/S, \nabla^{E})\right]_{[n]}(p).$$

The block diagonalization, the branch of the determinant square root, the Fourier factors, and the last symbol-supertrace identity are the standard Mehler calculation in [3, Chapter 4]; they are quoted here rather than concealed in an equality between raw and normalized curvatures. The formulas record exactly how its two factors become $\widehat{A}$ and the relative Chern character. □

The determinant factor is $\widehat{A}(TM)$, and the relative trace of the constant curvature exponential is $\mathrm{ch}(E/S)$.

5.3 The pointwise theorem

Theorem 5.2 (Local index theorem for Dirac operators)**.** *Let D be the Dirac operator of a graded Clifford module E over a closed even-dimensional Riemannian manifold. If $K(t;x,y)$ is the heat kernel of D^2, then*

$$\lim_{t\downarrow 0} \mathrm{str}_E\,K(t;x,x)\;d\,\mathrm{vol}(x) = \left[\widehat{A}(TM, \nabla^{TM})\,\mathrm{ch}(E/S, \nabla^{E})\right]_{[n]}(x).$$

Proof from the rescaled heat-kernel theorem. We use the standard rescaled heat-kernel theorem: in synchronous gauge, the Getzler-rescaled kernel of D^2 converges with all derivatives on compact sets to the kernel of $\mathcal{H}_p$, and the rescaled remainder has uniform Gaussian bounds. This is the analytic core of the local index theorem; complete constructions are given in [3, Chapters 4–5] and [4]. The ordinary heat parametrix by itself does not imply this uniform rescaled statement.

Given that theorem, Clifford cancellation shows that coefficients of Getzler degree below n have zero supertrace. After parabolic rescaling, the only surviving coefficient is the heat kernel of the curvature model. The model lemma and the Clifford top-degree supertrace identify it with the displayed Chern–Weil form.

For the remainder, choose enough terms in the heat parametrix that its error is $O(t^N)$ in every required C^k-seminorm. Under $x = \sqrt{t}\,X$, a term of Getzler order $n - j$ gains $t^{j/2}$. Terms killed by supertrace vanish identically, and every surviving term other than the model therefore tends to zero. The Volterra correction has the same estimate by the ordered-simplex bound from the heat-kernel construction. This proves smooth convergence of the local density, not merely convergence after integration. □

Question 5.3. *Why does the supertrace retain only the top exterior degree?*

Answer. Getzler rescaling turns the Clifford filtration into the exterior algebra. The fibrewise supertrace vanishes on every Clifford degree below n, as proved in the Clifford-cancellation lemma. Degree above n is zero in $\Lambda^\bullet T_p^*M$. Thus only degree n survives, and that degree is naturally an n-form that can be integrated over M. □

Integrating and applying McKean–Singer gives the global formula.

Theorem 5.4 (Dirac index theorem)**.** *Under the same hypotheses,*

$$\boxed{\operatorname{ind} D^+ = \int_M \left[\widehat{A}(TM)\,\mathrm{ch}(E/S)\right]_{[n]}.}$$

For a spin manifold and $E = S \otimes W$,

$$\operatorname{ind} D_W^+ = \int_M \left[\widehat{A}(TM)\,\mathrm{ch}(W)\right]_{[n]}.$$

For W trivial, this is the $\widehat{A}$-genus.

5.4 Metric and connection independence

The left side of the index formula is constant under continuous deformations through elliptic operators. On the right, the transgression formula shows that a change of connection alters each characteristic form by an exact form. Since M is closed, its integral is unchanged.

This agreement separates the two roles in the proof: heat-kernel analysis identifies a local curvature density, while Chern–Weil theory proves that its integral is topological.

Chapter 6

Gauss–Bonnet and the signature theorem

6.1 The Euler operator

On an oriented Riemannian manifold, the de Rham operator

$$D_{\mathrm{dR}} = d + d^*$$

is the Dirac operator on the Clifford module $\Lambda^\bullet T^*M_{\mathbb{C}}$, graded by parity.

Theorem 6.1 (Chern–Gauss–Bonnet). *If M is closed, oriented, and even-dimensional, then*

$$\chi(M) = \int_M e(TM) = \int_M \mathrm{Pf}\left(\frac{R^{TM}}{2\pi}\right).$$

Proof. Hodge theory identifies

$$\mathrm{ind}\, D^+_{\mathrm{dR}} = \sum_j (-1)^j \dim H^j(M;\mathbb{C}) = \chi(M).$$

Apply the local index theorem to $\Lambda^\bullet T^*M_{\mathbb{C}}$. The splitting principle reduces its characteristic form to a product over oriented two-plane summands. In each summand, the Clifford-module factor cancels the denominator in the $\widehat{A}$-form and leaves the curvature root. Their product is the Pfaffian $\mathrm{Pf}(R^{TM}/2\pi)$.

In detail, if $x_1, \ldots, x_m$ are the formal oriented curvature roots, then

$$\widehat{A}(TM) = \prod_{j=1}^m \frac{x_j/2}{\sinh(x_j/2)}.$$

The graded relative Chern character of the de Rham Clifford module is

$$\mathrm{ch}(\Lambda^\bullet/S) = \prod_{j=1}^m \left(e^{x_j/2} - e^{-x_j/2}\right) = \prod_{j=1}^m 2\sinh(x_j/2).$$

Their product is $\prod_j x_j$, which is the Euler form. □

If $\dim M$ is odd, Poincaré duality gives $\chi(M) = 0$, and there is no top-degree Euler form.

For a closed oriented surface,

$$e(TM) = \frac{K}{2\pi}\, dA,$$

where K is Gaussian curvature. Hence

$$\int_M K\, dA = 2\pi\chi(M),$$

the classical Gauss–Bonnet formula.

Example 6.2 (Surface of revolution)**.** This corrects the coordinate calculation discussed in the original manuscript. Parametrize the generating curve by arclength and write the induced metric as

$$g = ds^2 + f(s)^2 d\theta^2.$$

For the oriented orthonormal coframe $\vartheta^1 = ds$, $\vartheta^2 = f(s)d\theta$, Cartan's first structure equation

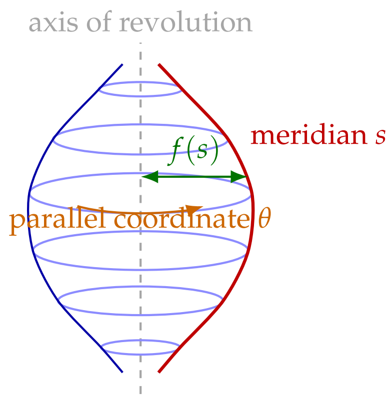

Figure 6.1: The arclength coordinate s follows a meridian, while θ runs around a parallel of radius $f(s)$. These orthogonal directions give $g = ds^2 + f(s)^2 d\theta^2$.

$$d\vartheta^i + \omega^i{}_j \wedge \vartheta^j = 0$$

gives

$$\omega^1{}_2 = -f'(s)d\theta, \qquad \omega^2{}_1 = f'(s)d\theta.$$

Since the structure group is $SO(2)$, the quadratic term in the curvature equation vanishes and

$$\Omega^1{}_2 = d\omega^1{}_2 = -f''(s)ds \wedge d\theta = -\frac{f''(s)}{f(s)}\vartheta^1 \wedge \vartheta^2.$$

Thus

$$K(s) = -\frac{f''(s)}{f(s)}.$$

This computation is frame-independent because a change of oriented orthonormal frame changes $\omega^1{}_2$ by an exact one-form and leaves $d\omega^1{}_2$ unchanged.

6.2 The signature operator

Let $\dim M = 4k$. On complex-valued p-forms define the chirality operator

$$\tau = i^{p(p-1)+2k} * .$$

Then

$$\tau^2 = 1, \qquad \tau D_{\mathrm{dR}} = -D_{\mathrm{dR}}\tau.$$

Let $\Omega^\pm$ be the ± 1-eigenspaces of τ. The signature operator is

$$D^+_{\mathrm{sign}} : \Omega^+(M) \longrightarrow \Omega^-(M), \qquad D^+_{\mathrm{sign}} = D_{\mathrm{dR}}|_{\Omega^+}.$$

Proposition 6.3.

$$\operatorname{ind} D^+_{\mathrm{sign}} = \operatorname{sign}(M),$$

the signature of the cup-product form on $H^{2k}(M;\mathbb{R})$.

Proof. The kernel of D_{dR} is the space of harmonic forms. For $p < 2k$, the involution τ exchanges $\mathcal{H}^p$ and $\mathcal{H}^{4k-p}$. On their direct sum it has matrix $\left(\begin{smallmatrix} 0 & U^{-1} \\ U & 0 \end{smallmatrix}\right)$ for an isomorphism U, so its $+1$ and -1 eigenspaces have equal dimension. In middle degree the phase $i^{p(p-1)+2k}$ equals one when $p = 2k$, so $\tau = *$. The positive and negative eigenspaces of $*$ have dimensions equal to the positive and negative indices of

$$([\alpha],[\beta]) \longmapsto \int_M \alpha \wedge \beta.$$

Their difference is the signature. $\square$

Theorem 6.4 (Hirzebruch signature theorem)**.** *For a closed oriented $4k$-manifold,*

$$\boxed{\operatorname{sign}(M) = \int_M L(TM).}$$

Proof. Apply the local index theorem to the Clifford module $(\Lambda^* T^* M_{\mathbb{C}}, c, \tau)$. The relative trace here is the one defined by the *signature* grading τ; it must not be replaced by the ordinary Chern character of the exterior algebra or by the even/odd grading used for Gauss–Bonnet.

Let $\pm x_1, \ldots, \pm x_{2k}$ be the formal normalized curvature roots of $T_{\mathbb{C}}M$. Diagonalize the creation and annihilation operators in each oriented curvature two-plane and take the τ-supertrace on their tensor product. The signature-module character identity is

$$\left[\widehat{A}(TM)\,\mathrm{ch}_\tau(\Lambda^* T^* M_{\mathbb{C}}/S)\right]_{[4k]} = \left[\prod_{j=1}^{2k} \frac{x_j}{\tanh x_j}\right]_{[4k]}. \tag{6.1}$$

The equality asserted is the top-degree identity needed by the index theorem; lower-degree normalizations depend on the ambient dimension and grading convention. In a two-plane calculation, the curvature action is diagonal on the vacuum and occupied states, while τ supplies the phase pairing complementary degrees. Multiplying those eigenvalue traces and the factor $(x_j/2)/\sinh(x_j/2)$ from $\widehat{A}$ gives the right-hand side of (6.1). The complete Clifford-character calculation, including the phase, is the standard signature example in [3, Chapter 4].

The product on the right is the Hirzebruch L-form by definition, and the analytic index is the signature by the preceding proposition. $\square$

In dimension four,

$$L(TM)_{[4]} = \frac{1}{3}p_1(TM), \qquad \operatorname{sign}(M) = \frac{1}{3}\int_M p_1(TM).$$

Chapter 7

Complex geometry and Riemann–Roch

7.1 Almost complex and complex manifolds

An almost complex structure on a $2m$-manifold is an endomorphism $J : TM \to TM$ with $J^2 = -1$. Complexifying the tangent bundle gives

$$T_{\mathbb{C}}M = T^{1,0}M \oplus T^{0,1}M,$$

the $+i$ and $-i$ eigenspaces of J. The projections are

$$\Pi^{1,0} = \frac{1}{2}(I - iJ), \qquad \Pi^{0,1} = \frac{1}{2}(I + iJ).$$

For a real vector v,

$$v^{1,0} = \frac{1}{2}(v - iJv), \qquad v^{0,1} = \frac{1}{2}(v + iJv).$$

Dualizing and taking exterior powers gives

$$\Lambda^k T^*_{\mathbb{C}}M = \bigoplus_{p+q=k} \Lambda^{p,q}T^*M.$$

$$\begin{tikzcd}
 & T^{1,0}M & \\
TM \arrow[r, "\mathbb{C}\otimes_{\mathbb{R}}(-)"] & T_{\mathbb{C}}M \arrow[u, "\Pi^{1,0}"'] \arrow[d, "\Pi^{0,1}"'] & \Lambda^k T^*_{\mathbb{C}}M = \bigoplus_{p+q=k} \Lambda^{p,q}T^*M. \\
 & T^{0,1}M &
\end{tikzcd}$$

Figure 7.1: An almost complex structure splits the complexified tangent bundle and induces the (p,q)-decomposition of forms.

Proposition 7.1. *Every almost complex manifold is canonically oriented.*

Proof. Choose real vectors $v_1, \dots, v_m$ such that

$$v_1, Jv_1, \dots, v_m, Jv_m$$

is a basis and declare it positive. A second complex frame is related to the first by a matrix $A \in \mathrm{GL}(m, \mathbb{C})$. Viewed as a real matrix,

$$\det_{\mathbb{R}} A = |\det_{\mathbb{C}} A|^2 > 0.$$

Thus every complex change of frame preserves the declared orientation, which patches smoothly over M. $\square$

The Nijenhuis tensor is

$$N_J(X,Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

Theorem 7.2 (Newlander–Nirenberg). *An almost complex structure is induced by holomorphic coordinate charts if and only if $N_J = 0$.*

Discussion. Newlander–Nirenberg is a nonlinear local existence theorem and is used here as a standard input. The necessity is elementary: coordinate vector fields of type $(1,0)$ are closed under brackets, which forces $N_J = 0$. The converse solves a first-order nonlinear system for holomorphic coordinates and is not part of the index-theoretic proof.

Such a structure is called integrable, and M is then a complex manifold. On a complex manifold,

$$d = \partial + \bar\partial,$$

with

$$\partial : \Omega^{p,q} \to \Omega^{p+1,q}, \qquad \bar\partial : \Omega^{p,q} \to \Omega^{p,q+1},$$

and $d^2 = 0$ implies

$$\partial^2 = 0, \qquad \bar\partial^2 = 0, \qquad \partial\bar\partial + \bar\partial\partial = 0.$$

On a nonintegrable almost complex manifold, d also has components of bidegrees $(2,-1)$ and $(-1,2)$. In particular, the projected $\bar\partial$-operator does not generally square to zero. This distinction is essential: the canonical spin^c Dirac operator still exists, but there need not be a Dolbeault complex.

Question 7.3. *What exactly does integrability change?*

Answer. An almost complex structure always splits complexified forms by type and always determines a canonical spin^c symbol. Integrability is the extra condition that the $(0,1)$-part of d squares to zero. Only then does $\bar\partial$ define cohomology and only then can the spin^c index be interpreted as a holomorphic Euler characteristic. $\square$

Example 7.4 (Complex projective space). The charts

$$U_j = \{[z_0 : \cdots : z_m] : z_j \neq 0\}, \qquad [z] \longmapsto \left(\frac{z_0}{z_j}, \dots, \frac{\widehat{z_j}}{z_j}, \dots, \frac{z_m}{z_j}\right)$$

have holomorphic transition functions and define the complex manifold $\mathbb{CP}^m$. Its induced almost complex structure is integrable by construction, and the Fubini–Study form makes it Kähler.

7.2 Almost Hermitian and almost Kähler geometry

An almost Hermitian structure is a pair (J, g) satisfying

$$g(JX, JY) = g(X, Y).$$

Its fundamental form

$$\omega(X, Y) = g(JX, Y)$$

is a nondegenerate real two-form. Compatible metrics always exist.

Question 7.5. *Does defining an almost complex structure require a metric?*

Answer. No. The datum is only the bundle endomorphism J with $J^2 = -I$, and it already determines the type decomposition and canonical orientation. A metric is introduced afterward. Averaging an arbitrary metric with its J-transform produces a compatible one, as the next proposition shows; there is no canonical choice in general. □

Proposition 7.6. *Every almost complex structure admits a compatible Riemannian metric.*

Proof. Start with any Riemannian metric g_0 and set

$$g(X, Y) = \frac{1}{2}\big(g_0(X, Y) + g_0(JX, JY)\big).$$

This is positive definite, and

$$g(JX, JY) = \frac{1}{2}\big(g_0(JX, JY) + g_0(J^2X, J^2Y)\big) = g(X, Y).$$

□

An almost Hermitian manifold is *almost Kähler* when

$$d\omega = 0.$$

It is *Kähler* when J is integrable and $d\omega = 0$. For an almost Hermitian structure,

$$\nabla^{TM} J = 0 \quad \Longleftrightarrow \quad N_J = 0 \text{ and } d\omega = 0.$$

Thus parallel J characterizes the Kähler condition, not the almost Kähler condition alone.

$$\begin{tikzcd}
 & \nabla^{TM} J = 0 \arrow[d, leftrightarrow] & \\
\text{almost Kähler} & \text{Kähler} \arrow[l, "d\omega=0"'] \arrow[r, "N_J=0"] & \text{complex Hermitian}
\end{tikzcd}$$

Figure 7.2: Kähler geometry is the simultaneous symplectic and integrable specialization of almost Hermitian geometry.

Discussion. The two lower conditions are independent in general. A symplectic manifold with a compatible almost complex structure is almost Kähler, but the chosen J need not be integrable. Conversely, a Hermitian metric on a complex manifold need not have closed fundamental form. Kähler geometry is the intersection of the symplectic and integrable conditions.

The Hermitian metric induces an inner product on (p,q)-forms. The complex-linear extension of the Hodge star has type

$$* : \Omega^{p,q}(M) \longrightarrow \Omega^{m-q,m-p}(M).$$

On a compact complex Hermitian manifold, define

$$\Box_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial.$$

This is elliptic, and Hodge theory gives

$$H^{p,q}_{\bar\partial}(M) \cong \ker\big(\Box_{\bar\partial} : \Omega^{p,q} \to \Omega^{p,q}\big).$$

Theorem 7.7 (Kähler identities)**.** *Let $L = \omega \wedge (-)$ and let $\Lambda = L^*$. Then*

$$[\Lambda, \partial] = i\bar\partial^*, \qquad [\Lambda, \bar\partial] = -i\partial^*.$$

Consequently,

$$\Box_\partial = \Box_{\bar\partial}, \qquad \Delta_d = 2\Box_{\bar\partial}.$$

Proof. Fix $p \in M$, choose a unitary Kähler-normal coframe $\{\zeta^j\}$, and apply both sides to an arbitrary test form α. At p, the connection coefficients vanish and $\nabla\omega = 0$, so

$$\partial\alpha = \sum_j \varepsilon(\zeta^j)\nabla_{Z_j}\alpha, \qquad \bar\partial\alpha = \sum_j \varepsilon(\bar\zeta^j)\nabla_{\bar Z_j}\alpha,$$

whereas

$$\partial^*\alpha = -\sum_j \iota_{Z_j}\nabla_{\bar Z_j}\alpha, \qquad \bar\partial^*\alpha = -\sum_j \iota_{\bar Z_j}\nabla_{Z_j}\alpha.$$

The algebraic relation

$$\iota_v\varepsilon(\alpha) + \varepsilon(\alpha)\iota_v = \alpha(v)$$

gives

$$[\Lambda, \varepsilon(\zeta^j)] = -i\iota_{\bar Z_j}, \qquad [\Lambda, \varepsilon(\bar\zeta^j)] = i\iota_{Z_j}.$$

Since $\nabla\Lambda = 0$, substitution in the preceding first-order formulas yields

$$[\Lambda, \partial]\alpha = i\bar\partial^*\alpha, \qquad [\Lambda, \bar\partial]\alpha = -i\partial^*\alpha.$$

The point and test form were arbitrary, so these are equalities of first-order differential operators; no tensoriality assertion about those operators is needed.

The graded Jacobi identity and $\partial\bar\partial + \bar\partial\partial = 0$ give

$$\Box_{\bar\partial} = -i[\bar\partial, [\Lambda, \partial]] = i[\partial, [\Lambda, \bar\partial]] = \Box_\partial.$$

The same identities cancel the mixed terms in Δ_d, leaving $\Box_\partial + \Box_{\bar\partial} = 2\Box_{\bar\partial}$. □

It follows that

$$H^k(M; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar\partial}(M)$$

on every compact Kähler manifold.

7.3 The almost Dolbeault operator

On an almost complex manifold the type decomposition exists, but the exterior derivative has four components:

$$d = \mu + \partial + \bar{\partial} + \bar{\mu},$$

of bidegrees

$$(2,-1), \qquad (1,0), \qquad (0,1), \qquad (-1,2).$$

The components μ and $\bar{\mu}$ are algebraic expressions in the Nijenhuis tensor and vanish exactly when J is integrable. Thus $\bar{\partial}$ exists by projection for every almost complex structure, but $\bar{\partial}^2$ need not vanish.

Nevertheless,

$$S = \Lambda^{0,\bullet} T^* M, \qquad S^{\pm} = \Lambda^{0,\mathrm{ev}/\mathrm{odd}} T^* M$$

is the spinor bundle of the canonical spin^c structure. On a real covector ξ, Clifford multiplication is

$$c(\xi) = \sqrt{2}\left(\varepsilon(\xi^{0,1}) - \iota_{(\xi^\sharp)^{0,1}}\right), \qquad c(\xi)^2 = -|\xi|^2.$$

Choose a Hermitian connection preserving J and g. It induces a Clifford connection on S, hence a Dirac operator

$$D_J : C^\infty(S^+) \longrightarrow C^\infty(S^-).$$

Its principal symbol is $i\,c(\xi)$, so D_J is elliptic. The operator $\sqrt{2}(\bar{\partial} + \bar{\partial}^*)$ has the same principal symbol and differs from D_J by a zero-order torsion term. Therefore the two operators have the same index even when $\bar{\partial}$ is not a differential.

Theorem 7.8 (Almost-complex Riemann–Roch). *Let M^{2m} be compact almost complex and let E be a complex vector bundle. For the canonical spin^c Dirac operator twisted by E,*

$$\operatorname{ind} D^+_{J,E} = \int_M \left[\operatorname{Td}(T^{1,0}M)\operatorname{ch}(E)\right]_{[2m]}.$$

Proof. The determinant line of the canonical spin^c structure is

$$K_M^{-1} = \det T^{1,0}M.$$

The spin^c index theorem gives

$$\operatorname{ind} D^+_{J,E} = \int_M \left[e^{c_1(K_M^{-1})/2}\widehat{A}(TM)\operatorname{ch}(E)\right]_{[2m]}.$$

If $x_1, \ldots, x_m$ are the formal Chern roots of $T^{1,0}M$, then

$$e^{x_j/2}\frac{x_j/2}{\sinh(x_j/2)} = \frac{x_j}{1-e^{-x_j}}.$$

Multiplying these identities yields

$$e^{c_1(K_M^{-1})/2}\widehat{A}(TM) = \operatorname{Td}(T^{1,0}M),$$

which proves the formula. □

Question 7.9. *What is lost when the almost complex structure is not integrable?*

Answer. Not the elliptic symbol and not its index. What is lost is the cochain complex: $\bar{\partial}^2$ may be nonzero, so there is no Dolbeault cohomology whose alternating dimension equals the index. Integrability turns the same spin^c index into the holomorphic Euler characteristic. □

7.4 Holomorphic vector bundles

A holomorphic structure on a complex vector bundle $E \to M$ is an operator

$$\bar{\partial}_E : \Omega^{0,q}(E) \longrightarrow \Omega^{0,q+1}(E)$$

satisfying the Leibniz rule and

$$\bar{\partial}_E^2 = 0.$$

It defines the Dolbeault complex

$$0 \longrightarrow \Omega^{0,0}(E) \xrightarrow{\bar{\partial}_E} \Omega^{0,1}(E) \longrightarrow \cdots \longrightarrow \Omega^{0,m}(E) \longrightarrow 0.$$

Its cohomology is denoted $H^{0,q}(M;E)$, or $H^q(M,\mathcal{O}(E))$.

Proposition 7.10. *The Dolbeault complex is elliptic.*

Proof. At a nonzero real covector ξ, the principal symbol of $\bar{\partial}_E$ is exterior multiplication by $i\xi^{0,1}$. The symbol complex is therefore the Koszul complex

$$\Lambda^{0,\bullet} T_x^* M \otimes E_x \xrightarrow{\xi^{0,1}\wedge} \Lambda^{0,\bullet+1} T_x^* M \otimes E_x.$$

Contraction by the dual of $\xi^{0,1}$, divided by $|\xi^{0,1}|^2$, is a contracting homotopy. Hence the symbol complex is exact.

Indeed, if

$$h_\xi = \frac{\iota_{(\xi^{0,1})^\sharp}}{|\xi^{0,1}|^2},$$

then exterior multiplication and contraction satisfy

$$h_\xi(\xi^{0,1}\wedge) + (\xi^{0,1}\wedge)h_\xi = I.$$

For a nonzero real covector ξ, its $(0,1)$-part cannot vanish: complex conjugation would also make its $(1,0)$-part vanish. Thus the homotopy is defined at every point of $T^*M \setminus 0$. $\square$

Given a Hermitian metric on a holomorphic bundle, there is a unique unitary connection whose $(0,1)$-part is $\bar{\partial}_E$. It is the Chern connection. Its curvature has type $(1,1)$.

In a holomorphic frame with Hermitian metric matrix h, the Chern connection has

$$\nabla^{0,1} = \bar{\partial}, \qquad \nabla^{1,0} = \partial + h^{-1}\partial h,$$

with the placement of h^{-1} determined by the column-vector convention. Metric compatibility gives this formula, hence uniqueness. Since the $(0,1)$-part is integrable and the $(2,0)$-part cancels by the Maurer–Cartan identity, the curvature is of type $(1,1)$.

The Dolbeault–Dirac operator is

$$D_E = \sqrt{2}\,(\bar{\partial}_E + \bar{\partial}_E^*).$$

It acts from even to odd antiholomorphic degree and has the principal symbol of the canonical spin^c Dirac operator twisted by E. On a Kähler manifold the operators agree; in general they differ, for standard connection choices, by a zero-order torsion term and have the same index. Hodge theory gives

$$\operatorname{ind} D_E^+ = \sum_{q=0}^{m} (-1)^q \dim H^{0,q}(M;E) =: \chi(M,E).$$

The logical chain is

$$
\begin{tikzcd}
(\Omega^{0,\bullet}(E), \bar\partial_E) \arrow[r, "\text{Hodge theory}"] & \ker D_E^{\pm} \arrow[r, "\text{graded dimension}"] & \operatorname{ind} D_E^{+} \arrow[d, equal] \\
H^{0,\bullet}(M;E) \arrow[u, "\cong"] \arrow[rr, "\sum(-1)^q \dim"'] & & \chi(M,E).
\end{tikzcd}
$$

Figure 7.3: For an integrable complex structure, Dolbeault Hodge theory identifies the Dirac index with the holomorphic Euler characteristic.

7.5 Hirzebruch–Riemann–Roch

The canonical spin^c structure of an almost complex manifold has determinant line K_M^{-1}, where

$$K_M = \Lambda^{m,0} T^* M.$$

The spin^c Dirac index formula is

$$\operatorname{ind} D_E^+ = \int_M \left[e^{c_1(K_M^{-1})/2} \widehat{A}(TM)\,\mathrm{ch}(E) \right]_{[2m]}.$$

For a complex vector bundle V,

$$\mathrm{Td}(V) = e^{c_1(V)/2} \widehat{A}(V_{\mathbb{R}}).$$

Taking $V = T^{1,0}M$ yields the Riemann–Roch density.

This identity is checked one Chern root at a time:

$$e^{x/2} \frac{x/2}{\sinh(x/2)} = \frac{x}{1 - e^{-x}}.$$

The splitting principle then proves it for every complex vector bundle.

Theorem 7.11 (Hirzebruch–Riemann–Roch). *If M is a compact complex manifold and E is a holomorphic vector bundle, then*

$$\boxed{\chi(M,E) = \int_M \left[\mathrm{Td}(T^{1,0}M)\,\mathrm{ch}(E) \right]_{[2m]}.}$$

Proof. The analytic index of the Dolbeault–Dirac operator is $\chi(M,E)$ by Dolbeault Hodge theory. It has the same principal symbol as the canonical spin^c Dirac operator twisted by E; their difference is zero-order, so their indices agree. Apply the spin^c Dirac index theorem and use

$$c_1(K_M^{-1}) = c_1(T^{1,0}M), \qquad e^{c_1/2} \widehat{A}(TM) = \mathrm{Td}(T^{1,0}M).$$

□

The same characteristic number is defined for a compact almost complex manifold and equals the index of its canonical spin^c Dirac operator. Integrability is needed only to identify that index with holomorphic sheaf cohomology.

7.6 Riemann surfaces

Let X be a compact connected Riemann surface of genus g, and let $L \to X$ be a holomorphic line bundle.

7.6.1 Divisors, meromorphic sections, and degree

A divisor is a finite formal sum

$$D = \sum_{p \in X} n_p[p], \qquad n_p \in \mathbb{Z},$$

and $\deg D = \sum_p n_p$. If f is meromorphic, its principal divisor is

$$(f) = \sum_p \operatorname{ord}_p(f)[p].$$

The argument principle on a compact Riemann surface gives $\deg(f) = 0$. Consequently degree descends from divisors to divisor classes.

The line bundle $\mathcal{O}(D)$ is characterized locally as follows. After trivializing by its distinguished meromorphic section, a *holomorphic* section is represented by a meromorphic function that is allowed a pole of order at most n_p if $n_p > 0$, and is required to vanish to order at least $-n_p$ if $n_p < 0$. Equivalently,

$$H^0(X, \mathcal{O}(D)) = \{f \text{ meromorphic} : (f) + D \geq 0\} \cup \{0\}.$$

Proposition 7.12 (Divisor–line-bundle correspondence)**.** *Every divisor determines a holomorphic line bundle $\mathcal{O}(D)$ with a distinguished meromorphic section s_D satisfying $(s_D) = D$. Conversely, a nonzero meromorphic section s of a holomorphic line bundle L determines a divisor (s) and an isomorphism*

$$L \cong \mathcal{O}((s)).$$

Two divisors define isomorphic line bundles precisely when they differ by a principal divisor.

Proof. Choose disks U_p around the support of D and a complementary open set U_0. On $U_p \cap U_0$, use the transition function $z_p^{-n_p}$. These functions form a holomorphic cocycle and define $\mathcal{O}(D)$; the local functions $z_p^{n_p}$ and 1 patch to a meromorphic section with divisor D. Conversely, local expressions $s = f_i e_i$ in holomorphic frames have orders that agree on overlaps, because the transition functions are nowhere zero. Dividing the local frame by the corresponding power of a coordinate identifies L with $\mathcal{O}((s))$. Replacing s by fs changes its divisor by (f), and the same local construction proves the last assertion. □

Proposition 7.13. *For a holomorphic line bundle on a compact Riemann surface,*

$$\deg L = \deg(s) = \int_X c_1(L)$$

for every nonzero meromorphic section s.

Proof. Choose disjoint coordinate disks around the zeros and poles of s, and use s as a nonvanishing holomorphic frame on their complement. On a disk about p, choose a nonvanishing holomorphic frame e_p and write

$$s = f_p e_p, \qquad f_p = z^{m_p} g_p, \qquad g_p(0) \neq 0.$$

The first Chern number is the sum of the winding numbers of these clutching functions. With the complex boundary orientation,

$$\int_X c_1(L) = \sum_p \frac{1}{2\pi i} \int_{|z|=\varepsilon} d\log f_p = \sum_p m_p.$$

The factor g_p has winding number zero. Thus the Chern number equals the divisor degree, independently of the chosen meromorphic section. □

In particular, the divisor of a meromorphic one-form represents the canonical bundle K_X. Since $c_1(K_X) = -c_1(T^{1,0}X)$, Gauss–Bonnet gives $\deg K_X = -(2-2g) = 2g-2$.

The degree of L is therefore

$$\deg L = \int_X c_1(L).$$

Since only degree-two terms contribute,

$$\mathrm{ch}(L) = 1 + c_1(L), \qquad \mathrm{Td}(T^{1,0}X) = 1 + \frac{1}{2}c_1(T^{1,0}X).$$

Gauss–Bonnet gives

$$\int_X c_1(T^{1,0}X) = \chi(X) = 2-2g.$$

Hirzebruch–Riemann–Roch therefore becomes

$$\chi(X,L) = \deg L + 1 - g.$$

Theorem 7.14 (Serre duality for a curve)**.** *There is a natural perfect pairing*

$$H^1(X,L) \times H^0(X, K_X \otimes L^{-1}) \longrightarrow \mathbb{C}, \qquad ([\alpha], s) \longmapsto \int_X \alpha \wedge s.$$

Hence

$$H^1(X,L) \cong H^0(X, K_X \otimes L^{-1})^*.$$

Proof. First, the pairing is well defined on cohomology. If α is replaced by $\alpha + \bar\partial_E u$ and s is holomorphic, then

$$\int_X (\bar\partial_E u) \wedge s = \int_X \bar\partial(u \wedge s) = 0$$

by Stokes' theorem.

Dolbeault theory identifies $H^1(X,L)$ with harmonic L-valued $(0,1)$-forms. Define the conjugate-linear Hermitian star

$$\bar{*}_L : \Omega^{0,1}(L) \longrightarrow \Omega^{1,0}(L^*) = \Gamma(K_X \otimes L^{-1})$$

by

$$\beta \wedge \bar{*}_L \alpha = \langle \beta, \alpha \rangle \, d\,\mathrm{vol} \qquad (\beta, \alpha \in \Omega^{0,1}(L)),$$

with the chosen complex orientation. Its formal-adjoint identity is

$$\bar\partial_{K_X \otimes L^{-1}}(\bar{*}_L \alpha) = -\bar{*}_L(\bar\partial_L^* \alpha).$$

Thus a harmonic α determines the holomorphic section $s_\alpha = \bar{*}_L \alpha$. Conversely, $\bar{*}_L^{-1}$ sends a holomorphic section to a top-degree $\bar\partial$-closed form annihilated by $\bar\partial_L^*$, hence to a harmonic form. Finally,

$$\int_X \alpha \wedge s_\alpha = \|\alpha\|_{L^2}^2.$$

Every nonzero harmonic class therefore has a nonzero pairing, and the inverse construction proves nondegeneracy in the other factor. This proves the perfect pairing and the stated duality. □

Combining Serre duality with the index formula gives the classical Riemann–Roch theorem.

Theorem 7.15 (Riemann–Roch for a line bundle)**.**

$$\boxed{h^0(X, L) - h^0(X, K_X \otimes L^{-1}) = \deg L + 1 - g.}$$

Proof. Dolbeault theory gives

$$\chi(X, L) = h^0(X, L) - h^1(X, L).$$

Serre duality identifies

$$h^1(X, L) = h^0(X, K_X \otimes L^{-1}),$$

while Hirzebruch–Riemann–Roch in complex dimension one gives

$$\chi(X, L) = \int_X \left(c_1(L) + \frac{1}{2}c_1(T^{1,0}X)\right) = \deg L + 1 - g.$$

Combining the three equalities proves the formula. □

If a holomorphic line bundle has a nonzero holomorphic section, its degree is the number of zeros of that section counted with multiplicity and is therefore nonnegative. Thus, if $\deg L > 2g - 2$, then

$$\deg(K_X \otimes L^{-1}) < 0$$

and the second term vanishes:

$$h^0(X, L) = \deg L + 1 - g.$$

Discussion. The last vanishing uses only a simple divisor argument. A nonzero holomorphic section s has isolated zeros, and the local orders of vanishing sum to $\deg L$. Hence a negative-degree line bundle cannot have a nonzero holomorphic section. Since $\deg K_X = 2g - 2$, the bundle $K_X \otimes L^{-1}$ has negative degree under the displayed hypothesis.

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